



```

name: SimulationBaltic
log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
log type: smcl
opened on: 25 Aug 2026, 18:54:44

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
> tion of Benefits for Older People in the Baltic States up to 2036 with data from SHA
> RE and Eurostat) fs(12) save(report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
2. clear all
3. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
> 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
4. use "sharew`num'_rel9-0-0_gv_imputations.dta"
5. keep if implicat==1
6. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
7. isid mergeid
8. drop _merge
9. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
10. isid mergeid
11. drop _merge
12.
8 . if `num'==8{
13. gen Qyear=2020
14. }
15. if `num'==9{
16. gen Qyear=2022
17. }
18.
9 . keep if country==57 | country==48 | country==35
19.
10. save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
> \W`num'_final.dta", replace
20. }
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
> 0 ALL_datasets_stata
(214,780 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

Result Number of obs
Not matched 0
Matched 53,695 (_merge==3)

(label country already defined)
(label language already defined)

```

Result	Number of obs
Not matched	0
Matched	53,695

(47,632 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W8_final.dta saved

C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-

> 0 ALL datasets stata

(277,788 observations deleted)

(label **orienti** already defined)

(label **eurod** already defined)

(label **numeracy** already defined)

(label **gali** already defined)

(label **language** already defined)

(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	69,447

(label **country** already defined)

(label **language** already defined)

Result	Number of obs
Not matched	0
Matched	69,447

(61,831 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W9_final.dta saved

11.

12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	5,101

16. drop _merge

17. save "W8_final.dta", replace
file W8_final.dta saved

18.

19. clear all

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
21. use "W9_final.dta"
```

```
22. merge 1:1 mergeid using "weightslong.dta", keep(match)
    (label country already defined)
```

Result	Number of obs	
Not matched	0	
Matched	4,921	(_merge ==3)

```
23. drop _merge
```

```
24. save "W9_final.dta", replace
    file W9_final.dta saved
```

```
25.
```

```
26. clear all
```

```
27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
28. use "W8_final.dta"
```

```
29. append using "W9_final.dta"
    (variable psu was str10, now str11 to accommodate using data's values)
    (variable ssu was str6, now str13 to accommodate using data's values)
    (label pstat already defined)
    (label language already defined)
    (label country already defined)
    (label Hrespt already defined)
    (label pcskill already defined)
    (label IMPflag already defined)
    (label orienti already defined)
    (label bfi10 already defined)
    (label thospital already defined)
    (label satisfied already defined)
    (label often already defined)
    (label yedu already defined)
    (label fdistress already defined)
    (label writing already defined)
    (label reading already defined)
    (label hearing already defined)
    (label eyesight already defined)
    (label mstat already defined)
    (label sphus already defined)
    (label noyes already defined)
    (label hnrsc already defined)
    (label undersq already defined)
    (label clarif already defined)
    (label willans already defined)
    (label otrf already defined)
    (label empstat already defined)
    (label cjs already defined)
    (label isced already defined)
    (label gender already defined)
    (label num already defined)
    (label gali already defined)
    (label meat already defined)
    (label legeggs already defined)
    (label fruit already defined)
    (label numeracy already defined)
    (label numeracy2 already defined)
    (label eurod already defined)
    (label dairyp already defined)
    (label weekday already defined)
    (label specday already defined)
    (label dkrf already defined)
    (label dummi already defined)
```

```

(label zero_one already defined)
(label loneli already defined)
(label chronic2 already defined)
(label sphus2 already defined)
(label bmi2 already defined)
(label mobil2 already defined)
(label mobil3 already defined)
(label adl2 already defined)
(label iadl2 already defined)
(label phactiv already defined)

30.
31. save "dataset.dta", replace
    file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. label define isc 0 "None" 1 "Primary education" 2 "Lower secondary education" 3 "Upper
    > secondary education" 4 "Post-secondary non-tertiary education" 5 "Bachelor's or e
    > quivalent level" 6 "Master's or equivalent level/Doctoral or equivalent level"

40. label values isced isc

41. label variable isced "Education"

42.
43. generate income = thinc-ypen3

44.
45. egen id= group(mergeid)

46.
47. drop if cjs== -99
    (70 observations deleted)

48.
49. generate educ=0

50. replace educ=1 if isced==3 | isced==4
    (5,234 real changes made)

51. replace educ=2 if isced>=5
    (2,943 real changes made)

52. label define educL 0 "Less than primary, primary and lower secondary education (ISCE
    > D levels 0-2)" 1 "Upper secondary and post-secondary non-tertiary education (ISCED l
    > evels 3 and 4)" 2 "Tertiary education (ISCED levels 5-8)"

53. label values educ educL

54.
55. keep single ypen3 gender cjs age educ sphus country income gali id Qyear sphus

```

```

56.
57. drop if ypen3>7000
   (7 observations deleted)

58.
59. *additional age categorization to match Eurostat categories
60. generate ageD=0

61. replace ageD=1 if age>=55 & age<=64
   (2,789 real changes made)

62. replace ageD=2 if age>=65 & age<=74
   (3,129 real changes made)

63. replace ageD=3 if age>=75 & age<=84
   (2,530 real changes made)

64. replace ageD=4 if age>=85
   (848 real changes made)

65. label define ageDl 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
   > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

66. label values ageD ageDl

67.
68. generate ageE=0

69. replace ageE=1 if age>=55 & age<=64
   (2,789 real changes made)

70. replace ageE=2 if age>=65
   (6,507 real changes made)

71. label define age1 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
   > r than 64 years"

72. label values ageE age1

73.
74. *>>>DESCRIPTIVES
75.
76. tabulate country

```

Country identifier	Freq.	Percent	Cum.
Estonia	5,520	55.51	55.51
Lithuania	2,422	24.35	79.86
Latvia	2,003	20.14	100.00
Total	9,945	100.00	

```

77.
78. asdoc codebook single ypen3 gender cjs age educ sphus country income gali id Qyear,
   > compact save(report.doc) append

```

Variable	Obs	Unique	Mean	Min	Max	Label
single	9945	2	.4113625	0	1	Single
ypen3	9945	378	232.8604	0	6816	Disability pensions/benefits
gender	9945	2	1.631875	1	2	Gender
cjs	9945	6	1.949321	1	97	Current job situation
age	9945	64	69.68678	34	100	Age
educ	9945	3	1.117748	0	2	
sphus	9945	5	3.740875	1	5	Self-perceived health - US scale
country	9945	3	42.59698	35	57	Country identifier
income	9945	4450	10096.15	-.0000737	170698	
gali	9945	2	.5882353	0	1	Limitation with activities
id	9945	5086	2559.401	1	5101	group(mergeid)

Qyear 9945 2 2020.979 2020 2022

Click to Open File: [report.doc](#)

79.

80. asdoc codebook, save(report.doc) append

country	Country identifier
----------------	---------------------------

Type: Numeric (**byte**)
 Label: **country**

Range: [35,57] Units: 1
 Unique values: 3 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	5,520	35	Estonia
	2,422	48	Lithuania
	2,003	57	Latvia

single	Single
---------------	---------------

Type: Numeric (**byte**)
 Label: **noyes**

Range: [0,1] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	5,854	0	No
	4,091	1	Yes

ypen3	Disability pensions/benefits
--------------	-------------------------------------

Type: Numeric (**double**)

Range: [0,6816] Units: 1.000e-07
 Unique values: 378 Missing .: 0/9,945

Mean: 232.86
 Std. dev.: 884.674

Percentiles:	10%	25%	50%	75%	90%
	0	0	0	0	276

gender	Gender
---------------	---------------

Type: Numeric (**byte**)
 Label: **gender**

Range: [1,2] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	3,661	1	Male
	6,284	2	Female

age	Age
------------	------------

Type: Numeric (**int**)
 Label: **num**, but **64** nonmissing values are not labeled

Range: [34,100] Units: 1
 Unique values: 64 Missing .: 0/9,945

Examples: 60
 66
 72
 80

sphus **Self-perceived health - US scale**

Type: Numeric (**byte**)
 Label: **sphus**

Range: [1,5] Units: 1
 Unique values: 5 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	154	1	Excellent
	416	2	Very good
	2,842	3	Good
	4,974	4	Fair
	1,559	5	Poor

gali **Limitation with activities**

Type: Numeric (**byte**)
 Label: **gali**

Range: [0,1] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	4,095	0	Not limited
	5,850	1	Limited

cjs **Current job situation**

Type: Numeric (**byte**)
 Label: **cjs**

Range: [1,97] Units: 1
 Unique values: 6 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	6,180	1	Retired
	2,947	2	Employed or self-employed
	241	3	Unemployed
	436	4	Permanently sick
	96	5	Homemaker
	45	97	Other

Qyear **(unlabeled)**

Type: Numeric (**float**)

Range: [2020,2022] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation: Freq. Value
 5,075 2020
 4,870 2022

income (unlabeled)

Type: Numeric (**float**)

Range: [-.00007374,170697.95] Units: 1.000e-20
 Unique values: 4,450 Missing .: 0/9,945

Mean: 10096.1
 Std. dev.: 8970.94

Percentiles: 10% 25% 50% 75% 90%
 2000 4920 7800 12898.5 20000

id group (mergeid)

Type: Numeric (**float**)

Range: [1,5101] Units: 1
 Unique values: 5,086 Missing .: 0/9,945

Mean: 2559.4
 Std. dev.: 1474.54

Percentiles: 10% 25% 50% 75% 90%
 512 1281 2565 3838 4597

educ (unlabeled)

Type: Numeric (**float**)
 Label: **educL**

Range: [0,2] Units: 1
 Unique values: 3 Missing .: 0/9,945

Tabulation: Freq. Numeric Label
 1,772 0 Less than primary, primary and
 lower secondary education
 (ISCED levels 0-2)
 5,230 1 Upper secondary and
 post-secondary non-tertiary
 education (ISCED levels 3 and
 4)
 2,943 2 Tertiary education (ISCED
 levels 5-8)

ageD (unlabeled)

Type: Numeric (**float**)
 Label: **ageD1**

Range: [0,4] Units: 1
 Unique values: 5 Missing .: 0/9,945

Tabulation: Freq. Numeric Label
 649 0 Less than 55 years old
 2,789 1 Between 55 and 64 years old
 3,129 2 Between 65 and 74 years old
 2,530 3 Between 75 and 84 years old
 848 4 Older than 85 years

90.

```
91. asdoc tabstat income if country==35 & gender==1, by(ageD) stat(mean), save(report.do
> c) append
aged
```

	income
Between_55~d	17976.43
Between_65~d	15632.26
Between_75~d	14499.49
Less_than_~d	12239.24
Older_than~s	10595.08

Click to Open File: [report.doc](#)

```
92. asdoc tabstat income if country==48 & gender==1, by(ageD) stat(mean), save(report.do
> c) append
aged
```

	income
Between_55~d	8339.341
Between_65~d	9607.151
Between_75~d	8643.868
Less_than_~d	8286.833
Older_than~s	6938.143

Click to Open File: [report.doc](#)

```
93. asdoc tabstat income if country==57 & gender==1, by(ageD) stat(mean), save(report.do
> c) append
aged
```

	income
Between_55~d	4309.18
Between_65~d	6241.763
Between_75~d	8744.98
Less_than_~d	8120.891
Older_than~s	7283.539

Click to Open File: [report.doc](#)

```
94. asdoc tabstat income if country==35 & gender==2, by(ageD) stat(mean), save(report.do
> c) append
aged
```

	income
Between_55~d	17266.16
Between_65~d	14092.54
Between_75~d	11699.28
Less_than_~d	8704.869
Older_than~s	7086.693

Click to Open File: [report.doc](#)

```
95. asdoc tabstat income if country==48 & gender==2, by(ageD) stat(mean), save(report.do
> c) append
aged
```

	income
Between_55~d	7885.404
Between_65~d	8610.249
Between_75~d	7541.797
Less_than_~d	6174.841
Older_than~s	4534.537

Click to Open File: [report.doc](#)

```

96. asdoc tabstat income if country==57 & gender==2, by(ageD) stat(mean), save(report.do
> c) append
ageD

```

	income
Between_55~d	6621.115
Between_65~d	5830.542
Between_75~d	7075.679
Less_than~d	5807.162
Older_than~s	4390.64

Click to Open File: [report.doc](#)

```

97.

```

```

98. *>>>ANALYSIS

```

```

99.

```

```

100 xtset id Qyear

```

Panel variable: **id** (unbalanced)

Time variable: **Qyear**, 2020 to 2022, but with gaps

Delta: **1 unit**

```

101

```

```

102 *-----

```

```

103 *ESTONIA

```

```

104

```

```

105 *general model

```

```

106

```

```

107 xttobit ypen3 i.gender age i.gali i.educ income i.single i.cjs if country==35, ll(0)

```

```

> iterate(25) baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12371.283
Iteration 1: Log likelihood = -8541.0028
Iteration 2: Log likelihood = -8047.2479
Iteration 3: Log likelihood = -8005.607
Iteration 4: Log likelihood = -8005.1643
Iteration 5: Log likelihood = -8005.1641

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -43929.156
Iteration 1: Log likelihood = -43908.934
Iteration 2: Log likelihood = -43908.866
Iteration 3: Log likelihood = -43908.866

```

Fitting full model:

```

Iteration 0: Log likelihood = -11891.02 (not concave)
Iteration 1: Log likelihood = -9201.2195 (not concave)
Iteration 2: Log likelihood = -8354.7229
Iteration 3: Log likelihood = -8018.0066
Iteration 4: Log likelihood = -7937.3718
Iteration 5: Log likelihood = -7918.9839
Iteration 6: Log likelihood = -7918.8372
Iteration 7: Log likelihood = -7918.837

```

Random-effects tobit regression

Limits: Lower = 0
Upper = +infGroup variable: id
Random effects u_i ~ GaussianNumber of obs = 5,520
Uncensored = 761
Left-censored = 4,759
Right-censored = 0Number of groups = 2,837
Obs per group: min = 1
avg = 1.9
max = 2

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -7918.837

Wald chi2(12) = 854.40
Prob > chi2 = 0.0000

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0	(base)						
> 166.5082	136.731	1.22	0.223	-101.4796	434.4961	Female	
						age	
> -78.3567	8.991099	-8.71	0.000	-95.97893	-60.73447		
						gali	
						Not limited	
> 0	(base)					Limited	
> 2352.885	167.8879	14.01	0.000	2023.831	2681.94		
						educ	
						Less than primary, primary and lower secondary education (ISCED levels 0-2)	
> 0	(base)						
						Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)	-
> 337.3164	165.6163	-2.04	0.042	-661.9185	-12.71441		
						Tertiary education (ISCED levels 5-8)	-
> 889.3742							

>	202.3882							
>		-4.39						
>			0.000					
>				-1286.048				
>					-492.7007			
						income		-
>	.0393951							
>		.0085828						
>		-4.59						
>			0.000					
>				-.0562172				
>					-.0225731			
						single		
						No		
>	0							
>	(base)					Yes		
>	17.74069							
>		149.0098						
>			0.12					
>			0.905					
>				-274.3132				
>					309.7945			
						cjs		
						Retired		
>	0							
>	(base)					Employed or self-employed		
>	334.858							
>		194.3659						
>			1.72					
>			0.085					
>				-46.09223				
>					715.8082			
						Unemployed		
>	1213.511							
>		399.8693						
>			3.03					
>			0.002					
>				429.7813				
>					1997.24			
						Permanently sick		
>	4445.234							
>		247.3918						
>			17.97					
>			0.000					
>				3960.355				
>					4930.113			
						Homemaker		-
>	466.7972							
>		809.0859						
>		-0.58						
>			0.564					
>				-2052.576				
>					1118.982			
						Other		-
>	1020.291							
>		958.281						
>		-1.06						
>			0.287					
>				-2898.487				
>					857.9051			
						_cons		
>	1385.401							
>		702.8						
>			1.97					
>			0.049					
>				7.938				

```

>
2762.864
-----
/ sigma_u |
> 1721.842
>      89.95124
>      19.14
>      0.000
>      1545.541
>      1898.143
>      / sigma_e |
> 1747.443
>      63.01815
>      27.73
>      0.000
>      1623.93
>      1870.956
-----
rho |
> .4926211
>      .0325444
>      .4291996
>      .55623
-----

```

LR test of sigma_u=0: chibar2(01) = **172.65** Prob >= chibar2 = **0.000**

```

108 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(CoreEE) replace
  results.xls
  dir : seeout

```

```

109 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreEE) noaster replace
  resultsNoStar.xls
  dir : seeout

```

```

110
111 *for disability
112
113 xttobit ypen3 i.gender i.ageD i.gali income if country==35, ll(0) iterate(25) basele
> vels

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12574.16
Iteration 1: Log likelihood = -8690.0137
Iteration 2: Log likelihood = -8178.4506
Iteration 3: Log likelihood = -8140.6469
Iteration 4: Log likelihood = -8140.4696
Iteration 5: Log likelihood = -8140.4696

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -44691.727
Iteration 1: Log likelihood = -44643.617
Iteration 2: Log likelihood = -44642.958
Iteration 3: Log likelihood = -44642.958

```

Fitting full model:

```
Iteration 0: Log likelihood = -11743.477
Iteration 1: Log likelihood = -9291.477 (not concave)
Iteration 2: Log likelihood = -8334.8979
Iteration 3: Log likelihood = -8045.086
Iteration 4: Log likelihood = -8002.4898
Iteration 5: Log likelihood = -8001.4655
Iteration 6: Log likelihood = -8001.465
```

Random-effects tobit regression

Limits: Lower = 0
Upper = +inf

Group variable: **id**
Random effects $u_i \sim \text{Gaussian}$

Number of obs = 5,520
Uncensored = 761
Left-censored = 4,759
Right-censored = 0

Number of groups = 2,837
Obs per group:
min = 1
avg = 1.9
max = 2

Integration method: **mvaghermite**

Integration pts. = 12

Log likelihood = -8001.465

Wald chi2(7) = 589.33
Prob > chi2 = 0.0000

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
	Male	0 (base)				
	Female	2.055042	152.2603	0.01	0.989	-296.3696
> 300.4797						
	aged					
	Less than 55 years old	0 (base)				
	Between 55 and 64 years old	615.5205	239.6016	2.57	0.010	145.9099
> 1085.131						
	Between 65 and 74 years old	-2731.041	265.0717	-10.30	0.000	-3250.572
> -2211.51						
	Between 75 and 84 years old	-3003.733	271.8857	-11.05	0.000	-3536.619
> 2470.846						
	Older than 85 years	-2852.003	318.1848	-8.96	0.000	-3475.633
> 2228.372						
	gali					
	Not limited	0 (base)				
	Limited	3027.507	192.0651	15.76	0.000	2651.066
> 3403.948						
	income	-.0675669	.0081949	-8.25	0.000	-.0836286
> .0515053						
	_cons	-2913.865	316.7808	-9.20	0.000	-3534.744
> 2292.986						
	/sigma_u	2189.74	98.66261	22.19	0.000	1996.365
> 2383.116						
	/sigma_e	1720.259	65.252	26.36	0.000	1592.368
> 1848.151						
	rho	.6183654	.0279853			.5625044
> .6718615						

LR test of sigma_u=0: chibar2(01) = 278.01

Prob >= chibar2 = 0.000

```

114 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityEE) append
  results.xls
  dir : seeout

115 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityEE) noaster append
  resultsNoStar.xls
  dir : seeout

116
117 *for income
118
119 xttobit ypen3 i.gender i.ageE i.gali income if country==35, ll(0) iterate(25) basele
> vels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12580.252
Iteration 1: Log likelihood = -8695.1587
Iteration 2: Log likelihood = -8180.6133
Iteration 3: Log likelihood = -8142.2205
Iteration 4: Log likelihood = -8142.0359
Iteration 5: Log likelihood = -8142.0359

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -44693.866
Iteration 1: Log likelihood = -44648.652
Iteration 2: Log likelihood = -44648.098
Iteration 3: Log likelihood = -44648.098

```

Fitting full model:

```

Iteration 0: Log likelihood = -11741.375
Iteration 1: Log likelihood = -9275.1461 (not concave)
Iteration 2: Log likelihood = -8291.5574
Iteration 3: Log likelihood = -8027.616
Iteration 4: Log likelihood = -8002.5992
Iteration 5: Log likelihood = -8002.4501
Iteration 6: Log likelihood = -8002.4499

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: mvaghermite

Log likelihood = -8002.4499

```

Number of obs      = 5,520
      Uncensored   = 761
      Left-censored = 4,759
      Right-censored = 0

```

```

Number of groups   = 2,837
Obs per group:
      min = 1
      avg = 1.9
      max = 2

```

Integration pts. = 12

```

Wald chi2(5)      = 584.97
Prob > chi2       = 0.0000

```

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
	Male	0 (base)				
	Female	-5.136616	152.1665	-0.03	0.973	-303.3775
> 293.1042						
	ageE					
	Less than 55 years old	0 (base)				
	Between 55 and 64 years old	610.5346	239.6779	2.55	0.011	140.7745
> 1080.295						
	Older than 64 years	-2853.651	247.7542	-11.52	0.000	-3339.24
> 2368.062						
	gali					
	Not limited	0 (base)				
	Limited	3017.349	192.0138	15.71	0.000	2641.009
> 3393.689						
	income	-.0670237	.0081849	-8.19	0.000	-.0830659
> .0509816						
	_cons	-2914.57	317.085	-9.19	0.000	-3536.045
> 2293.095						
	/sigma_u	2194.624	98.71581	22.23	0.000	2001.145
> 2388.104						
	/sigma_e	1719.476	65.26921	26.34	0.000	1591.551
> 1847.401						
	rho	.6196311	.0279166			.563895
> .6729855						

LR test of sigma_u=0: chibar2(01) = 279.17

Prob >= chibar2 = 0.000

120 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(IncomeEE) append
results.xls
dir : seeout

121 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeEE) noaster append
resultsNoStar.xls
dir : seeout

122
123 *-----
124 *LITHUANIA
125
126 *general model
127
128 xttobit ypen3 i.gender age i.gali i.educ income i.single i.cjs if country==48, ll(0)
> iterate(25) baselevels tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

Fitting full model:

```
Iteration 0: Log likelihood = -4625.028
Iteration 1: Log likelihood = -2522.8718
Iteration 2: Log likelihood = -2358.3883
Iteration 3: Log likelihood = -2351.1517
Iteration 4: Log likelihood = -2351.0609
Iteration 5: Log likelihood = -2351.05
Iteration 6: Log likelihood = -2351.0479
Iteration 7: Log likelihood = -2351.0475
Iteration 8: Log likelihood = -2351.0474
Iteration 9: Log likelihood = -2351.0474
```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -19057.685
Iteration 1: Log likelihood = -19054.176
Iteration 2: Log likelihood = -19054.171
Iteration 3: Log likelihood = -19054.171
```

Fitting full model:

```
Iteration 0: Log likelihood = -4350.2922
Iteration 1: Log likelihood = -2339.745
Iteration 2: Log likelihood = -2317.4931
Iteration 3: Log likelihood = -2315.8524
Iteration 4: Log likelihood = -2315.8234
Iteration 5: Log likelihood = -2315.8184
Iteration 6: Log likelihood = -2315.8174
Iteration 7: Log likelihood = -2315.8172
Iteration 8: Log likelihood = -2315.8172
Iteration 9: Log likelihood = -2315.8171
```

Random-effects tobit regression

Limits: Lower = 0
Upper = +inf

Group variable: id
Random effects u_i ~ Gaussian

Integration method: mvaghermite

Log likelihood = -2315.8171

Number of obs = 2,422
Uncensored = 217
Left-censored = 2,205
Right-censored = 0

Number of groups = 1,235
Obs per group:
min = 1
avg = 2.0
max = 2

Integration pts. = 12

Wald chi2(12) = 236.92
Prob > chi2 = 0.0000

						ypen3	Co
> efficient							
> Std. err.							
> z							
> P> z							
> [95% con							
> f. interval]							
						gender	
						Male	
> 0							
> (base)							
> 430.2501						Female	-
> 315.368							
> -1.36							
> 0.172							
> -1048.36							
> 187.8599							


```

>      724.4564
>
>      0.12
>      0.906
>      -1334.124
>      1505.693
>      Permanently sick |
> 5434.891
>      476.011
>      11.42
>      0.000
>      4501.927
>      6367.856
>      Homemaker | -
> 12569.39
>      575992.1
>      -0.02
>      0.983
>      -1141493
>      1116354
>      Other |
> 1464.288
>      1542.501
>      0.95
>      0.342
>      -1558.959
>      4487.535
>
>      _cons | -
> 725.6779
>      1997.76
>      -0.36
>      0.716
>      -4641.216
>      3189.86
>
>
>      /sigma_u |
> 2415.657
>      209.062
>      11.55
>      0.000
>      2005.903
>      2825.411
>      /sigma_e |
> 2143.949
>      154.0062
>      13.92
>      0.000
>      1842.102
>      2445.795
>
>
>      rho |
> .5593795
>      .0539046
>      .4528738
>      .6617326
>

```

LR test of sigma_u=0: chibar2(01) = 70.46

Prob >= chibar2 = 0.000

```

129 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(CoreLT) append
  results.xls
  dir : seeout

130 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreLT) noaster append
  resultsNoStar.xls
  dir : seeout

131
132 *for disability
133
134 xttobit ypen3 i.gender i.ageD i.gali income if country==48, ll(0) iterate(25) basele
> vels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

```

Fitting full model:

```

Iteration 0: Log likelihood = -4764.1859
Iteration 1: Log likelihood = -2552.9555
Iteration 2: Log likelihood = -2472.9916
Iteration 3: Log likelihood = -2469.9219
Iteration 4: Log likelihood = -2469.9122
Iteration 5: Log likelihood = -2469.9122

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -19435.708
Iteration 1: Log likelihood = -19427.452
Iteration 2: Log likelihood = -19427.424

```

Fitting full model:

```

Iteration 0: Log likelihood = -4420.5636
Iteration 1: Log likelihood = -2834.3118 (not concave)
Iteration 2: Log likelihood = -2478.6459 (not concave)
Iteration 3: Log likelihood = -2460.1033 (not concave)
Iteration 4: Log likelihood = -2449.3058 (not concave)
Iteration 5: Log likelihood = -2441.1009
Iteration 6: Log likelihood = -2413.4743
Iteration 7: Log likelihood = -2410.5426
Iteration 8: Log likelihood = -2410.5383
Iteration 9: Log likelihood = -2410.5378
Iteration 10: Log likelihood = -2410.5375

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: mvaghermite

Log likelihood = -2410.5375

```

Number of obs      = 2,422
Uncensored         = 217
Left-censored      = 2,205
Right-censored     = 0

```

```

Number of groups   = 1,235
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration pts. = 12

```

Wald chi2(7)       = 122.77
Prob > chi2        = 0.0000

```

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i	
> nterval]							
	gender						
	Male	0 (base)					
	Female	-1265.573	399.3644	-3.17	0.002	-2048.312	-
> 482.8328							
	aged						
	Less than 55 years old	0 (base)					
	Between 55 and 64 years old	1042.583	769.3484	1.36	0.175	-465.312	
> 2550.478							
	Between 65 and 74 years old	-2581.948	870.7532	-2.97	0.003	-4288.593	-
> 875.3034							
	Between 75 and 84 years old	-4072.417	956.9911	-4.26	0.000	-5948.085	-
> 2196.749							
	Older than 85 years	-3496.48	1077.152	-3.25	0.001	-5607.66	-
> 1385.301							
	gali						
	Not limited	0 (base)					
	Limited	3051.978	411.8677	7.41	0.000	2244.732	
> 3859.224							
	income	-.111984	.0266827	-4.20	0.000	-.1642812	-
> .0596868							
	_cons	-5161.367	904.0224	-5.71	0.000	-6933.218	-
> 3389.515							
	/sigma_u	3612.105	296.3682	12.19	0.000	3031.234	
> 4192.976							
	/sigma_e	2225.215	167.7635	13.26	0.000	1896.405	
> 2554.025							
	rho	.724895	.0422206			.6366399	
> .801056							

LR test of sigma_u=0: chibar2(01) = 118.75

Prob >= chibar2 = 0.000

135 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(DisabilityLT) append
results.xls
dir : seeout

136 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLT) noaster append
resultsNoStar.xls
dir : seeout

137

138 *for income

139

140 xttoibit ypen3 i.gender i.ageE i.gali income if country==48, ll(0) iterate(25) basele
> vels tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

Fitting full model:

```
Iteration 0: Log likelihood = -4768.0345
Iteration 1: Log likelihood = -2554.6722
Iteration 2: Log likelihood = -2475.7377
Iteration 3: Log likelihood = -2472.5256
Iteration 4: Log likelihood = -2472.5174
Iteration 5: Log likelihood = -2472.5174
```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -19439.417
Iteration 1: Log likelihood = -19431.683
Iteration 2: Log likelihood = -19431.659
```

Fitting full model:

```
Iteration 0: Log likelihood = -4422.5114
Iteration 1: Log likelihood = -2861.7338 (not concave)
Iteration 2: Log likelihood = -2472.2208 (not concave)
Iteration 3: Log likelihood = -2455.7317 (not concave)
Iteration 4: Log likelihood = -2447.5253
Iteration 5: Log likelihood = -2414.3573
Iteration 6: Log likelihood = -2413.4271
Iteration 7: Log likelihood = -2413.4246
Iteration 8: Log likelihood = -2413.4245
Iteration 9: Log likelihood = -2413.4245
```

Random-effects tobit regression

```
Limits: Lower = 0
        Upper = +inf
```

```
Group variable: id
Random effects u_i ~ Gaussian
```

```
Number of obs      = 2,422
      Uncensored   = 217
      Left-censored = 2,205
      Right-censored = 0
```

```
Number of groups   = 1,235
Obs per group:
      min = 1
      avg = 2.0
      max = 2
```

Integration method: **mvaghermite**

```
Integration pts.   = 12
```

Log likelihood = **-2413.4245**

```
Wald chi2(5)       = 120.35
Prob > chi2        = 0.0000
```

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i	
> nterval]							
	gender						
	Male	0 (base)					
	Female	-1303.161	397.0635	-3.28	0.001	-2081.391	-
> 524.9304							
	ageE						
	Less than 55 years old	0 (base)					
	Between 55 and 64 years old	1045.154	775.3252	1.35	0.178	-474.4559	
> 2564.763							
	Older than 64 years	-3214.788	845.0069	-3.80	0.000	-4870.971	-
> 1558.605							
	gali						
	Not limited	0 (base)					
	Limited	3041.987	413.3066	7.36	0.000	2231.921	
> 3852.053							
	income	-.112629	.0268737	-4.19	0.000	-.1653005	-
> .0599575							
	_cons	-5163.735	908.4122	-5.68	0.000	-6944.19	
> -3383.28							

> 4167.066	/sigma_u	3595.056	291.8474	12.32	0.000	3023.046
	/sigma_e	2259.032	168.8215	13.38	0.000	1928.148
> .7932209	rho	.7169223	.0421156			.6292236

LR test of sigma_u=0: chibar2(01) = 118.19

Prob >= chibar2 = 0.000

141 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeLT) append
results.xls
dir : seeout

142 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLT) noaster append
resultsNoStar.xls
dir : seeout

143
144 *-----
145 *LATVIA
146
147 *general model
148
149 xttobit ypen3 i.gender age i.gali i.educ income i.single i.cjs if country==57, ll(0)
> iterate(25) baselevels tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

Fitting full model:

Iteration 0: Log likelihood = -3378.1792
Iteration 1: Log likelihood = -1527.9194
Iteration 2: Log likelihood = -1370.2257
Iteration 3: Log likelihood = -1357.6801
Iteration 4: Log likelihood = -1357.1173
Iteration 5: Log likelihood = -1357.033
Iteration 6: Log likelihood = -1357.0171
Iteration 7: Log likelihood = -1357.014
Iteration 8: Log likelihood = -1357.0133
Iteration 9: Log likelihood = -1357.0132
Iteration 10: Log likelihood = -1357.0131

Obtaining starting values for full model:

Iteration 0: Log likelihood = -15088.499
Iteration 1: Log likelihood = -15088.021
Iteration 2: Log likelihood = -15088.018
Iteration 3: Log likelihood = -15088.018

Fitting full model:

```
Iteration 0: Log likelihood = -3319.5307 (not concave)
Iteration 1: Log likelihood = -2332.3665 (not concave)
Iteration 2: Log likelihood = -1982.3558
Iteration 3: Log likelihood = -1890.0338 (backed up)
Iteration 4: Log likelihood = -1790.3092
Iteration 5: Log likelihood = -1377.8159
Iteration 6: Log likelihood = -1359.4048
Iteration 7: Log likelihood = -1353.2351
Iteration 8: Log likelihood = -1352.889
Iteration 9: Log likelihood = -1352.8726
Iteration 10: Log likelihood = -1352.8708
Iteration 11: Log likelihood = -1352.8707
Iteration 12: Log likelihood = -1352.8706
```

```
Limits: Lower = 0
        Upper = +inf
```

```

Number of obs      =    2,003
   Uncensored      =     132
   Left-censored   =    1,871
   Right-censored  =         0

```

```
Group variable: id
Random effects u_i ~ Gaussian
```

```
Number of groups = 1,014
Obs per group:   min = 1
                  avg = 2.0
                  max = 2
```

Integration method: **mvaghermite**

Integration pts. = 12

Log likelihood = -1352.8706

```
Wald chi2(12)      = 172.83
Prob > chi2        = 0.0000
```

[illegible]

							educ	
>	Less than primary, primary and lower secondary education (ISCED levels 0-2)							
>	0							
>	(base)							
>	Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)							
>	516.0529	470.6474	1.10	0.273	-406.3991	1438.505		
>								
>								
>								
>								
>	167.6026	612.8902	-0.27	0.784	-1368.845	1033.64		
>								
>								
>								
>								
>	.1043325	.0383366	-2.72	0.006	-.1794708	-.0291943	income	-
>								
>								
>								
>								
>	0						single	
>	(base)						No	
>								
>	56.45317	351.839	-0.16	0.873	-746.0449	633.1385	Yes	-
>								
>								
>								
>								
>	0						cjs	
>	(base)						Retired	
>								
>	-71.3182	467.9578	-0.15	0.879	-988.4985	845.8622	Employed or self-employed	
>								
>								
>								
>								
>	10210.18	208471	-0.05	0.961	-418805.8	398385.4	Unemployed	-
>								
>								
>								
>								
>	4240.327	510.2107	8.31	0.000	3240.333	5240.322	Permanently sick	
>								
>								
>								
>								
>	10065.22	408884.7	-0.02	0.980	-811464.6	791334.1	Homemaker	-
>								
>								
>								
>								

```

> 1563.53                                Other |
>      1398.216
>      1.12
>      0.263
>      -1176.922
>      4303.983
>
> 2040.357                                _cons |
>      1880.224
>      1.09
>      0.278
>      -1644.815
>      5725.529
+-----+
> 1323.95                                /sigma_u |
>      249.4362
>      5.31
>      0.000
>      835.0638
>      1812.836
>
> 2079.494                                /sigma_e |
>      186.6363
>      11.14
>      0.000
>      1713.694
>      2445.294
+-----+
> .2884323                                rho |
>      .0937032
>      .1365689
>      .4919915
+-----+
LR test of sigma_u=0: chibar2(01) = 8.29          Prob >= chibar2 = 0.002

```

```

150 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(Core) append
  results.xls
  dir : seeout

```

```

151 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(Core) noaster append
  resultsNoStar.xls
  dir : seeout

```

```

152
153 *for disability
154
155 xttobit ypen3 i.gender i.ageD i.gali income if country==57, ll(0) iterate(25) basele
> vels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```

Iteration 0: Log likelihood = -3508.1931
Iteration 1: Log likelihood = -1552.3332
Iteration 2: Log likelihood = -1458.2424
Iteration 3: Log likelihood = -1446.9305
Iteration 4: Log likelihood = -1446.4041
Iteration 5: Log likelihood = -1446.3416
Iteration 6: Log likelihood = -1446.332
Iteration 7: Log likelihood = -1446.3302
Iteration 8: Log likelihood = -1446.3299
Iteration 9: Log likelihood = -1446.3298
Iteration 10: Log likelihood = -1446.3298

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -15446.083
Iteration 1: Log likelihood = -15442.374
Iteration 2: Log likelihood = -15442.365

```

Fitting full model:

```

Iteration 0: Log likelihood = -3290.9604
Iteration 1: Log likelihood = -1568.1837 (not concave)
Iteration 2: Log likelihood = -1493.2462
Iteration 3: Log likelihood = -1462.4917
Iteration 4: Log likelihood = -1426.2089
Iteration 5: Log likelihood = -1422.5514
Iteration 6: Log likelihood = -1422.4905
Iteration 7: Log likelihood = -1422.4792
Iteration 8: Log likelihood = -1422.4769
Iteration 9: Log likelihood = -1422.4765
Iteration 10: Log likelihood = -1422.4764
Iteration 11: Log likelihood = -1422.4764

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

```

Number of obs      = 2,003
Uncensored         = 132
Left-censored      = 1,871
Right-censored     = 0

```

```

Number of groups   = 1,014
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -1422.4764

```

Wald chi2(7)      = 93.56
Prob > chi2       = 0.0000

```

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
	Male	0 (base)				
> 400.6069	Female	-373.2551	394.8348	-0.95	0.344	-1147.117
	ageD					
	Less than 55 years old	0 (base)				
> 1647.314	Between 55 and 64 years old	487.8286	591.5851	0.82	0.410	-671.6569
> 905.2272	Between 65 and 74 years old	-2326.25	725.025	-3.21	0.001	-3747.273 -
> 3339.917	Between 75 and 84 years old	-5799.906	1255.119	-4.62	0.000	-8259.895 -
> 845945.5	Older than 85 years	-15268.31	439402.9	-0.03	0.972	-876482.1
	gali					

	Not limited	0 (base)				
	Limited	3813.023	544.0046	7.01	0.000	2746.794
> 4879.253						
	income	-.2022976	.0434352	-4.66	0.000	-.2874291
> .1171661						-
	_cons	-5492.628	845.2343	-6.50	0.000	-7149.257
> -3836						
	/sigma_u	2561.528	278.8294	9.19	0.000	2015.032
> 3108.023						
	/sigma_e	2057.444	191.2273	10.76	0.000	1682.646
> 2432.243						
	rho	.6078487	.0657611			.4754034
> .7287797						

LR test of sigma_u=0: chibar2(01) = 47.71

Prob >= chibar2 = 0.000

```

156 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityLV) append
  results.xls
  dir : seeout
157 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLV) noaster append
  resultsNoStar.xls
  dir : seeout
158
159 *for income
160
161 xttobit ypen3 i.gender i.ageE i.gali income if country==57, ll(0) iterate(25) basele
> vels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```

Iteration 0: Log likelihood = -3512.8922
Iteration 1: Log likelihood = -1556.1683
Iteration 2: Log likelihood = -1465.8304
Iteration 3: Log likelihood = -1456.8357
Iteration 4: Log likelihood = -1456.7348
Iteration 5: Log likelihood = -1456.7347

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -15451.53
Iteration 1: Log likelihood = -15448.037
Iteration 2: Log likelihood = -15448.03

```

Fitting full model:

```

Iteration 0: Log likelihood = -3293.7749
Iteration 1: Log likelihood = -1558.793 (not concave)
Iteration 2: Log likelihood = -1486.1352
Iteration 3: Log likelihood = -1443.4774
Iteration 4: Log likelihood = -1433.9155
Iteration 5: Log likelihood = -1433.5011
Iteration 6: Log likelihood = -1433.4985
Iteration 7: Log likelihood = -1433.4985

```

Random-effects tobit regression

Limits: Lower = 0
Upper = +inf

Group variable: id
Random effects u_i ~ Gaussian

Number of obs = 2,003
Uncensored = 132
Left-censored = 1,871
Right-censored = 0

Number of groups = 1,014
Obs per group:
min = 1
avg = 2.0
max = 2

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -1433.4985

Wald chi2(5) = 97.84
Prob > chi2 = 0.0000

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
	Male	0 (base)				
> 338.2558	Female	-427.051	390.4698	-1.09	0.274	-1192.358
	ageE					
	Less than 55 years old	0 (base)				
> 1604.564	Between 55 and 64 years old	425.362	601.6447	0.71	0.480	-753.84
> 2085.631	Older than 64 years	-3527.965	735.898	-4.79	0.000	-4970.299 -
	gali					
	Not limited	0 (base)				
> 4880.909	Limited	3797.722	552.6566	6.87	0.000	2714.535
> .1131246	income	-.1995472	.044094	-4.53	0.000	-.2859697 -
> 3803.659	_cons	-5475.326	852.907	-6.42	0.000	-7146.993 -
> 3075.145	/sigma_u	2535.451	275.3591	9.21	0.000	1995.757
> 2541.072	/sigma_e	2154.076	197.4503	10.91	0.000	1767.081
> .7025843	rho	.5807904	.0653757			.4506512

LR test of sigma_u=0: chibar2(01) = 46.47

Prob >= chibar2 = 0.000

```
162 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeLV) append
results.xls
dir : seeout

163 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLV) noaster append
resultsNoStar.xls
dir : seeout

164
165 log close SimulationBaltic
      name: SimulationBaltic
      log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
      log type: smcl
      closed on: 25 Aug 2026, 18:56:19
```
